

Effect of Panel Dust on Solar Pump Discharge in Central Sudan

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Abstract: This study assessed the technical and economic performance of solar-powered irrigation pumps in twelve smallholder schemes of Gezira State. Pump discharge, operating hours and maintenance records were collected over two seasons. Solar pumps reduced seasonal energy costs by 61 percent compared with diesel pumps, while discharge fell during the dusty months of the dry season. The study recommends regular panel cleaning and shared maintenance arrangements among farmer groups.

Keywords: dust, photovoltaic pumps, Sudan

1. Introduction

Photovoltaic pumping has spread quickly in central Sudan.